

Software Requirements Specification (SRS) - Nexus Learning Lab

1. Introduction

The Nexus Learning Lab is a centralized platform that bridges the gap between professional portfolio showcasing (Machine Learning/Full-Stack) and an interactive educational tutoring system for students.

2. Technology Stack Recommendation

For this project, to ensure high performance and seamless integration with MySQL and XAMPP:

- * **Backend:** **Django (Python)**. It provides a robust authentication system (including built-in Password Reset), excellent ORM for MySQL integration, and easy AI/ML model deployment.

- * **Database:** **MySQL** (via XAMPP). It is reliable and perfectly compatible with Django's database backend.

- * **Frontend:** **React.js** (for dynamic dashboard/AI chatbot) combined with **Bootstrap 5** (for rapid, responsive UI development).

- * **Authentication:** Django's `django.contrib.auth` module for secure login, logout, and password recovery workflows.

3. System Features

3.1 Visitor/Portfolio Module

- * **Landing Page:** Hero section with Neural Network visualization.

- * **Project Showcase:** Category-based filters (ML, Web Dev, IoT).

- * **AI Chatbot:** Trained on project data to assist visitors.

3.2 Authentication Module

- * **Login/Register:** Role-based (Student/Teacher).

- * **Password Recovery:** Secure "Forgot Password" flow via email (SMTP integration).

3.3 Student Dashboard

- * **Class Hub:** Personalized access to Class 8, 9-10, or HSC content.

- * **Online Exam:** Timed quiz interface with real-time feedback.

- * **Performance Tracking:** Visualization of marks over time.

3.4 Teacher Dashboard

- * **Student Manager:** CRUD (Create, Read, Update, Delete) functionality for students.

- * **Exam Builder:** Interface to create questions and set exam times.
- * **AI Insights:** ML-driven performance reports.

4. Database Schema (MySQL)

- * `User``: (Username, Email, Role: Teacher/Student, Password)
- * `StudentProfile``: (User_ID, Class, Batch)
- * `Quiz``: (ID, Teacher_ID, Title, Date)
- * `Question``: (ID, Quiz_ID, Question_Text, Options, Correct_Answer)
- * `Result``: (ID, Student_ID, Quiz_ID, Score)

5. UI/UX Design Specifications

- * **Theme:** Dark Mode (Midnight Navy #0A192F).
- * **Components:** Glassmorphism cards for dashboards.
- * **Navigation:** Sticky Navbar with integrated Login/Portal links.